

Project - Computerized management systems for sustainable maritime transport adapted to hybrid-electric propulsion technologies

Main objective:

The main objective of this project is to increase the flexibility and efficiency of operations while reducing the environmental footprint of hybrid-electric propulsion (HEP) tugboats through the integration of computerized management systems.

Specific objectives:

1. Optimize the operational efficiency of connected ships
2. Reduce the environmental impact of hybrid-electric technologies
3. Improve the planning and coordination of maritime operations

A. Definition

Current projects mainly focus on large vessels or generic hybrid system models, whereas this project targets **real-life applications** with **complex dynamic constraints** (load variability, frequent maneuvers, short transient regimes).

The originality of this research lies in the following areas:

1. The development of an **Energy Management System (EMS)** specifically designed for hybrid tugboats, integrating **technical variables** (SOC, instantaneous power), **environmental variables** (weather, tide), and **operational variables** (mission type, towing duration);
2. The use of **adaptive and hybrid algorithms** (GA, A-ECMS) for **dynamic decision-making**;
3. The development of a **flexible integrated simulation platform** to test various technical configurations (mechanical, electrical, and electrochemical);
4. The integration of a **comprehensive techno-economic assessment**, considering life cycle costs, regulatory compliance, and logistical impacts.

B. Model Development

Based on this analysis, a **hybrid-electric propulsion model** will be developed by defining:

1. **Problem assumptions:**

- Operational conditions: operating time ranges;
- Mission profiles and battery charge/discharge cycles;
- Interactions between vessels and port infrastructures;
- Technical characteristics of propulsion systems and associated operating modes.

2. Design and mathematical formulation:

- Modeling of hybrid-electric systems through an appropriate mathematical approach;
- Acquisition and integration of technical data (fuel consumption, operation duration, distance traveled, developed power, location, etc.).

C. Model Validation

The model will be **tested and validated** using a **numerical simulation tool** allowing to:

1. Compare the obtained results with the performance of conventional tugboats/systems (diesel);
 2. Evaluate and estimate **GHG emissions** according to current national and international regulatory frameworks;
 3. Develop **prospective scenarios** based on different strategies:
- Massive adoption of electrification;
 - Integration of biofuels as an alternative energy source.

According to the literature, several tools have been used, including:

- **Matlab**, which has proven effective for this type of system modeling and optimization, covering multiple engineering fields such as mechanics, electrical, and electrochemistry — particularly through environments such as *Simulink* (for modeling) and *Optimization Toolbox* (for optimization);
- The **R** environment, initially designed for statistical analysis, which has evolved toward data processing, mathematical modeling, and graphical visualization;
- The **Python** language with its different libraries (Panda, etc), which offers robust capabilities in data analysis and modeling;
- **Typhoon HIL402**, used for real-time simulation of hybrid-electric vessels.

The selected tool here is **Python**, but a **comparative analysis of the advantages and limitations** will be conducted to justify this choice.

D. Results Interpretation

Once the simulations are completed, the results will be analyzed according to the initial assumptions and optimization objectives.

The evaluation will be structured around **four main criteria**, detailed below, and will include the identification and analysis of **performance indicators (KPIs)**:

1. Minimization of energy consumption

To maximize system efficiency, quantified using:

- Average consumption per mission hour (kWh/h);
- Overall energy efficiency (useful energy vs total energy consumed);
- Energy saved per mission (in % or kWh);
- Battery utilization ratio compared to that of the main diesel engines.

These KPIs will allow comparison of performance between different systems (conventional, hybrid, hybrid with EMS) in realistic mission scenarios.

2. Optimization of operations and logistical performance, measured through:

- Average mission time or per mission category (minutes, hours, days);
- Schedule compliance rate (%);
- Energy productivity index (kWh per tonne-nautical mile);
- Number of missions completed per day, week, month, or year;
- Tugboat utilization rate (%).

These indicators will assess the system's efficiency in resource optimization while measuring its impact on the **fluidity, precision, and efficiency** of maritime operations.

3. Reduction of GHG emissions

To meet environmental objectives:

- Calculation of CO₂-equivalent emissions per mission (kgCO₂e);
- Relative GHG reduction (%);
- Distribution of emissions by source (electric vs thermal).

The confirmation of performance gains and alignment with regulatory frameworks will be demonstrated through the **compliance rate with IMO standards**.

4. Reduction of technical downtimes and maintenance costs

To ensure system profitability:

- Number of unplanned shutdowns (count);
- Mean time between failures (MTBF – in hours, days, months, years);
- Maintenance cost per operating hour (\$/h);
- Preventive alert rate (%);
- Average battery SOC degradation (% per month or % per year).

The **long-term reliability and profitability** benefits of the proposed system will be demonstrated through these criteria.

The ultimate goal is to **reproduce and simulate the operation of a fleet of interconnected hybrid-electric tugboats**, connected to the various systems and equipment mentioned above.

These vessels would operate within a **port environment**, while also carrying out other types of operations within a **riverine area**.

Implementation steps:

1. Review the various documents provided, understand the mandate and requirements, and ask questions where clarifications are needed.
2. Clean the documents by removing unnecessary information, if required.
3. Enable communication between the different files according to the **provided interconnection diagram**, if necessary.
4. Extract results as previously described, through **graphs and geographical maps or location data**, in order to analyze their content.